Instruments & Systems
Valve Automation & Control Technology

ULTRASONIC LEVEL TRANSMITTER

INSTALLATION INSTRUCTIONS



INSTRUMENTS & SYSTEMS

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1. Overview

Thank you very much for your purchase.Our company Ultrasonic level meter!

This instrument contains a number of patented technologies, which are safe, clean, high precision, long service life, stability and reliability,Easy to install and maintain And other characteristics,It is suitable for acid, alkali, salt, anti-corrosion, high temperature and other fields.

This instrument can be connected to display tables or various DCS systems through 4~20mA or RS485 (Modbus protocol or other customized protocols) to provide real-time liquid level data for industrial automation operation.

This instrument has the following characteristics:

- The circuit design selects high-quality power supply modules from the power supply part, and the components select imported highly stable and reliable devices, which can completely replace the same type of imported instruments.
- The patented acoustic intelligent technology software can carry out intelligent echo analysis without any debugging and other special steps. This technology has the functions of dynamic thinking and dynamic analysis.
- The patented technology of sound wave intelligence owned by our company greatly improves the accuracy of the instrument, and the liquid level accuracy reaches 0.3%, which can resist various interference waves.
- This instrument is a non-contact instrument, which does not come into direct contact with liquid, so the failure rate is low. The instrument provides a variety of installation methods, and users can completely califer the instrument through this manual.
- All the input and output lines of the instrument have the protection functions of lightning protection and short circuit prevention.

2. Technical indicators

Measuring range:	0~15m (selected according to the actual measured mileage)
Blind spot:	0.35m~0.5m
Ranging accuracy:	0.3% (standard conditions)
Range measurement resolution:	1 mm
Pressure:	Below 4 atmospheres
The instrument shows:	Self-contained LCD display liquid level or spatial distance
Analog output:	4~20mA
Digital output:	RS485, Modbus
Agreement or customized agreement	
Power supply voltage:	DC24V/AC220V, built-in lightning protection device
Ambient temperature:	-20℃~ +60℃
Protection level:	IP65

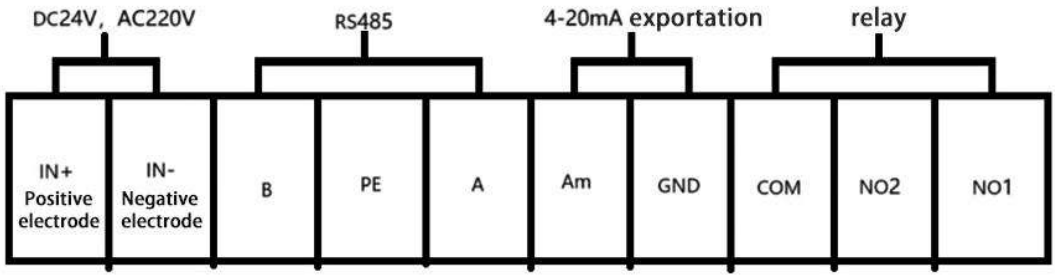
3. Instrument installation

3.1 Instrument shape Size (The size of the probe will change according to the range of the instrument. If there is a difference, we will inform you in advance)

Installation method:

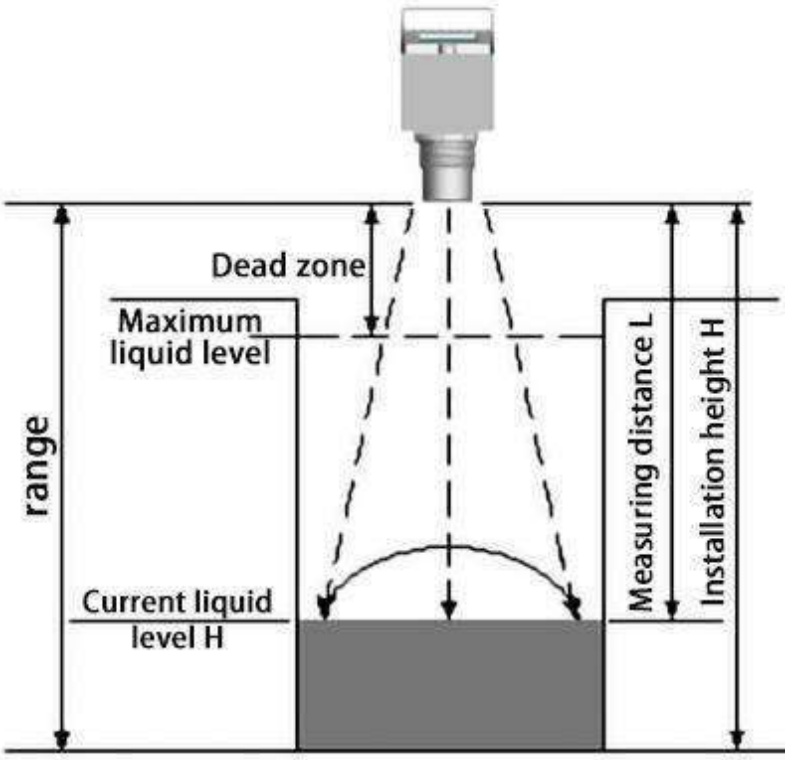
In an open environment, the bracket installation method is generally adopted, and the instrument is fixed with its own flange. Cut a round hole slightly larger than the probe diameter (60mm) in the installation position of the pool or tank, put the instrument in, and then tighten the flange from bottom to top. The installation must ensure that the probe surface of the instrument is at the level of the measured liquid level.

3.2 Instrument wiring: As shown in the figure below(This diagram is not universal. Please consult the manufacturer's wiring definition according to the specific model.)



Power supply: 220VAC or DC24V, inform when ordering; RS-485: A B(Second line 485 does not require PE port); Output signal: 4~20mA,Am is the output end, and GND is the ground line.

The two relays are respectively:Number1 is the first way, Number2 is the second way,COM is a public place.



3.3 The meaning of installation parameters

As shown in the figure above, the probe wave of the instrument is reflected back to the probe after hitting the liquid level. After the probe receives it, it calculates the time from the wave to the reception, and the measurement distance L is obtained. The instrument installation height TH minus the measurement distance L will obtain the current liquid level H.

The measuring range of the instrument refers to the distance that the instrument can measure, and the installation height TH should be less than the measuring range.

The blind area of the instrument refers to the area where the instrument cannot be measured near the probe. The distance between the highest liquid level and the probe should be larger than the blind area. If the blind area is 0.3m, the distance between the liquid level and the probe must be greater than 0.3m.

The probe wave is a diffusion process, that is, there is a direction angle. Pay attention to it when installing, otherwise it may hit the bulge of the pool wall or the edge of the channel.

3.4 Instrument installation principles

1) The distance from the emission surface of the probe to the lowest liquid level should be less than the range of the instrument.

2) The distance from the probe launch surface to the highest liquid level should be greater than the blind area of the instrument.

3) The emission surface of the probe should be parallel to the liquid surface.

4) The installation position of the probe should try to avoid the places where the liquid level fluctuates violently, such as the inlet and outlet directly below.

5) If the pool wall or tank wall is not smooth, the installation position of the instrument should be more than 0.3m away from the pool wall or tank wall.

6) If.Crane one's neckThe distance from the launch surface to the highest liquid level is less than the blind area of the instrument, and an extension tube needs to be installed.Extension pipe diameter

Greater than 120mm, length 0.35m~0.50m, vertical installation, smooth inner wall, the hole in the tank should be larger than the extension pipe.

Path. Or pass the pipe to the bottom of the tank, the pipe diameter is greater than 80mm, and the hole at the bottom of the pipe keeps the liquid level in the extension pipe equal to the height in the tank.

3.5 Installation Precautions

1) When the instrument is installed outdoors, it is recommended to install sunshades to extend the service life of the instrument. The installation is not less than the side wall of the container.1.Rice.

2) Wire and cable protection pipes, pay attention to sealing to prevent water accumulation.

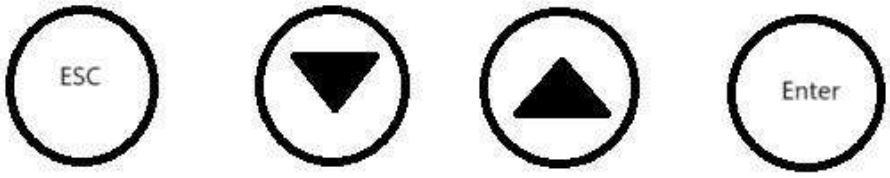
3) Although the instrument has its own lightning protection device, when the instrument is used in the multi-mine area, it is recommended to use it at the entry and exit line of the instrument separately.

Install a special lightning protection device.

4) When the instrument is used in particularly hot and cold places, that is, when the ambient temperature may exceed the working requirements of the instrument, it is recommended to install high and low temperature defense devices around the liquid level meter.

4. Instrument debugging

4.1 Kevboard Instructions



[**Esc**]**:** Leave Key.Leave Menu or Cancel the data change Of the time Leave Key.

[▼]: Turn down the key. In the menu, this key is used as the down flip key of the menu. When changing data, the key is used as an addition key.(Single-bit loop).

[▲]: Flip key and number key. In the menu, the key is used as the up flip key of the menu. When changing the data, the key does Shift Use the key.

[**Enter**]Confirm Key.Press [**Enter**] Enter the menu and press [**Enter**] Confirm Set up

For Installation Height / 4-20 mA Setting

Press **[Enter]** ➡ Password “1000" Press **[Enter]** ➡ User Setup Press **[Enter]** ➡ InstallationH Press **[Enter]** ➡ For Digit Shifting Press [»] and Press [⇕] ➡ Press **[Enter]** to save the range ➡ Press **[ESC]** to back ➡ Press **[⇕]** ➡ choose current setup menu Press **[Enter]** ➡ 20mAHeight Press **[Enter]** ➡ reset the Required Range and then Press **[ESC]** to back to the main menu

4.2 Parameter Settings

4.2.1System Settings and Advanced Settings Menu

System settings and advanced settings menus usually do not require user settings, otherwise it will affect the measurement and the data will be disordered!

If you need to change it, you need to set it under the guidance of the manufacturer! (The factory has been set up!)

4.2.2 Use Set up

When the instrument is working normally, press Enter Enter Set up Owner Menu,Use flipping up or flipping down Key selection Make the indicator "->" stay in the usage settings, press Enter Key to enter Enter Settings.

4.2.2.1The height of the probe (Installation height)

The probe can be displayed.The installation of Height(The total height of the probe to the bottom), can be viewed Dip one's hand into something Whether the head height conforms to the on-site conditions can also be used for liquid level calibration. Change the probe height value directly according to the on-site situation.

4.2..2.2 Output mode(Current output mode)

Sure Select the current output basis according to the actual demand, that is, the output corresponds to the liquid level height or the corresponding air height.

4.2.2.3 Communication Settings(Natal address and baud rate settings)

Sure Reset the address and baud rate of this machine according to actual needs. Please do not set or plug in or unplug the serial line when the instrument interacts with the upper computer (hot machine state)!You'd better restart the instrument after the setting is completed! (The upper computer needs to make corresponding changes to the parameter changes of the instrument)

RS485 (Modbus protocol format) sample command line (hexadecimal):

01 03 00 00 00 03 05 CB

Command interpretation (in order from the back):

01 is the instrument address (it can be changed according to the actual address, and the factory default is 01);

03 is the executed reading command (function code), and the user does not need to modify it and enter it according to the example;

00 00 00 03 is the high and low bits of the starting address of the read register and the high and low bits of the length of the register in order. This part of the user does not need to modify and enter according to the example;

05 CB is the 16-bit CRC check code at the front of the low byte (this instrument supports the 16-bit CRC check at the low byte at the top);

The data format received by RS485 (Modbus protocol format) (hexadecimal, the letter is the data temporarily represented):

01 03 06 aa aa bb bb cc cc xx xx

Interpretation of data format (from the back in order):

01 is the instrument address (it will change according to the actual address, and the factory default is 01);

03 is the executed reading command (function code);

06 is the length of the data read (this instrument reads the liquid level, air height and temperature at one time);

The aa aa data is liquid level data (16-bit unsigned integer data is divided into two 8-bit unsigned integer data, with high bytes in front), and the unit after converting decimal numbers is millimeters;

Bb bb data is empty high data (16-bit unsigned formatted data is divided into two 8-bit unsigned integer data, with high bytes first), and the unit after converting decimal numbers is millimeters;

Cc cc data is temperature (16-bit signed integer data is split into two 8-bit unsigned integer data, high bytes are in front, pay attention to the number of signed temperature bits, and it needs to be calculated according to the signed integer data after conversion!). After converting the decimal number, the data needs to be divided by 10 as the actual temperature, and the unit is Celsius;

Xx xx is the verified CRC verification code (this instrument supports low-byte CRC verification in the first 16 bits);

4.2.2.4 Current Parameter settings

4mA height and 20mA height settings You can choose Change the liquid level height corresponding to 4mA and 20mA according to the actual situation..

4mA coefficient and 20mA coefficient The setting is related to the current output calibration. Please set it under the guidance of the manufacturer.

4.2.2.5 Relay parameter settings

You can choose "&", "|", "N", "∧"Four kinds of symbols. The OK key at the time of change flashes the desired modification symbol, click the ▼ key to select the desired symbol, and the digital modification is the same as before.

The meaning of the symbol:

<: Less than the symbol

>: Greater than the symbol

&: and, indicating that both conditions need to be met;

|: Or, it means that one of the two conditions can be met;

N: Only the former condition, and the latter is not displayed;

∧: The former condition is the relay adsorption condition (generally used to open the pump), and the latter condition is the relay.

The opening condition (generally used to stop the pump) is mainly used for drainage well and inlet well control.

4.3 Display settings

4.3.1 Display mode

You can choose Text mode and echo mode.

4.3.2 Language selection

This machine supports English display modes. Sure Through this setting Choose The language displayed.

Remarks: You only need to set P02 (20mA setting) and P03 (TH installation height) for instrument debugging!

Ultrasonic level meter Warranty card instructions

Product model		**Product number**	
Acceptance date		**The person in charge of installation**	

Warranty policy:

- Users should show the warranty card during the repair. Appear due to normal use during the warranty period The fault of,You can enjoy the free warranty with the warranty card.
- Warranty period:Our company The product warranty period is from Be dispatched from the factory Count ten from the date Two.Within a month.

The following situations Be out Free warranty Within the scope:

- The product or its parts have exceeded the free warranty period.
- Because the use environment is not suitable for production Hard ware failure caused by product use requirements.
- Failure or damage caused by poor power supply environment or foreign matter entering the equipment.
- Faults caused by failure to operate according to the usage method and precautions written in the user manual.
- Failures caused by irresistible natural factors such as lightning, water fire and so on.
- Failure or damage caused by unauthorized dismantling and repair or unauthorized modification or abuse of the machine.

Restriction description:

- Please keep the warranty card as a warranty certificate. If it is lost, it will not be compensated.
- This warranty card has the authority to explain Root Owned by the company,Root The company has the right to modify the content of this card without prior notice.

SERVICE SUPPORT

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